HW # 1

1.3

(a) use the result

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi} \quad ; \quad \text{proof: } I = \int_{-\infty}^{\infty} e^{-x^2}\,dx$$

$$\text{also} \rightarrow \int_{-\infty}^{\infty} e^{-y^2}\,dy = I$$

$$\text{thus} \rightarrow I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-x^2} e^{-y^2}\,dx\,dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy.$$

Now work in cylindrical coordinates:

$$x^2+y^2 = r^2$$

$$dx\,dy = r\,dr\,d\theta$$

and limits of integration

$$\int_0^{\infty} dr \int_0^{2\pi} d\theta$$

$$\Rightarrow I^2 = \int_0^{\infty} r\,dr\,e^{-r^2} \int_0^{2\pi} d\theta$$

$$= \int_0^{\infty} \frac{dr^2 e^{-r^2}}{2} \int_0^{2\pi} d\theta$$

$$= \left(\frac{1}{2}\right) 2\pi = \pi$$

$$\Rightarrow I = \sqrt{\pi}$$

thus

(a) $$\int_{-\infty}^{\infty} \rho(x)\, dx = 1$$

$$\Rightarrow A \int_{-\infty}^{\infty} e^{-\lambda(x-a)^2} dx = 1$$

Choose $(x-a)\sqrt{\lambda} = y$

$$\Rightarrow dx = \frac{dy}{\sqrt{\lambda}}$$

$$\frac{A}{\sqrt{\lambda}} \int_{-\infty}^{\infty} e^{-y^2} dy = 1$$

$$A \sqrt{\frac{\pi}{\lambda}} = 1 \Rightarrow A = \sqrt{\frac{\lambda}{\pi}}$$

(b) $$\langle x \rangle = A \int_{-\infty}^{\infty} x\, e^{-\lambda(x-a)^2} dx$$

$$= A\Big[\underbrace{\int_{-\infty}^{\infty} (x-a) e^{-\lambda(x-a)^2} dx}_{\text{odd about } x \equiv a} \to 0 + a \int_{-\infty}^{\infty} e^{-\lambda(x-a)^2} dx\Big]$$

$$= a\, A \int_{-\infty}^{\infty} e^{-\lambda(x-a)^2} dx = a.$$

$$\langle x^2 \rangle = A \int_{-\infty}^{\infty} (x-a+a)^2 e^{-\lambda(x-a)^2} dx$$

$$= A \int_{-\infty}^{\infty} \Big[(x-a)^2 + \underbrace{2a(x-a)}_{\text{odd about } x=a} \to 0 + a^2\Big] e^{-\lambda(x-a)^2} dx$$

$$= A \int_{-\infty}^{\infty} (x-a)^2 e^{-\lambda(x-a)^2} dx + a^2 A \int_{-\infty}^{\infty} e^{-\lambda(x-a)^2} dx$$

$$\langle x^2 \rangle = A \int_{-\infty}^{\infty} (x-a)^2 e^{-\lambda(x-a)^2} dx + a^2 .$$

<u>let us evaluate</u> : $$J_1 = \int_{-\infty}^{\infty} (x-a)^2 e^{-\lambda(x-a)^2} dx$$

$$= \frac{1}{\lambda\sqrt{\lambda}} \int_{-\infty}^{\infty} y^2 e^{-y^2} dy$$

use → $$I_1[a] = \int_{-\infty}^{\infty} e^{-a y^2} dy = \sqrt{\pi/a}$$

$$-\frac{dI_1[a]}{da} = \int_{-\infty}^{\infty} y^2 e^{-ay^2} dy$$

$$\Rightarrow J_1 = \frac{1}{\lambda\sqrt{\lambda}} \left[-\frac{dI_1[a]}{da} \right] \Bigg|_{a=1}$$

$$= \frac{1}{\lambda\sqrt{\lambda}} \left[-\frac{d}{da} \sqrt{\frac{\pi}{a}} \right] \Bigg|_{a=1}$$

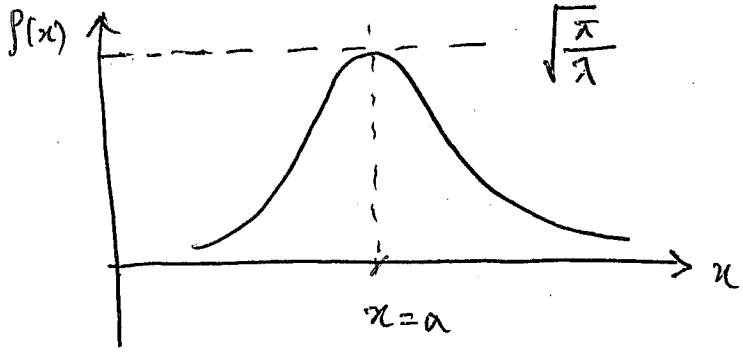
$$= \frac{1}{\lambda\sqrt{\lambda}} \frac{\sqrt{\pi}}{2a^{3/2}} \Bigg|_{a=1} = \frac{\sqrt{\pi}}{2\lambda\sqrt{\lambda}}$$

$$\Rightarrow \langle x^2 \rangle = A \frac{\sqrt{\pi}}{2\lambda\sqrt{\lambda}} + a^2 = \frac{1}{2\lambda} + a^2$$

$$\Rightarrow \sigma = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$$

$$= \sqrt{\frac{1}{2\lambda} + a^2 - a^2} = \sqrt{\frac{1}{2\lambda}}$$

(c)



$\boxed{1.4}$

$$\psi(x,0) = \begin{cases} A\,\dfrac{x}{a} & \text{if } 0 \le x \le a \\ A\,\dfrac{(b-x)}{b-a} & \text{if } a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

(a) $$\int_{-\infty}^{\infty} \psi^*(x)\,\psi(x)\,dx = 1$$

$$\Rightarrow \left[\int_{-\infty}^{0} + \int_{0}^{a} + \int_{a}^{b} + \int_{b}^{\infty} \right] \psi^*(x)\,\psi(x)\,dx = 1$$

Since $\psi(x) = 0$ in the region $\rightarrow -\infty \le x \le 0$ and $b \le x < \infty$

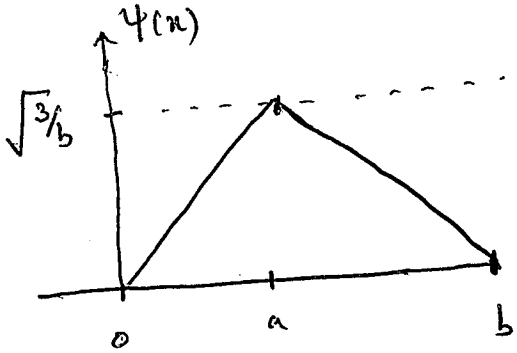
we have: $$\int_{0}^{a} \psi^*(x)\,\psi(x)\,dx + \int_{a}^{b} \psi^*(x)\,\psi(x)\,dx = 1$$

$$\Rightarrow \quad A^2 \left[\int_0^a \frac{x^2}{a^2}\, dx + \int_a^b \frac{(b-x)^2}{(b-a)^2}\, dx \right] = 1$$

$$\Rightarrow \quad A^2 \left[\frac{a}{3} + \left(-\frac{a}{3} + \frac{b}{3} \right) \right] = 1$$

$$\Rightarrow \quad A = \sqrt{\frac{3}{b}}$$

(b)



(c) the probability density $|\psi(x)|^2$ is highest at $\boxed{x = a}$

(d) left of $a \rightarrow$ $\quad P_{(-\infty, a)} = \int_{-\infty}^{a} |\psi(x)|^2 dx$

$$= \int_0^a A^2 \frac{x^2}{a^2}\, dx = \left(\frac{3}{b}\right) \frac{a}{3} = \frac{a}{b}$$

for $b = a \Rightarrow P_{(-\infty, a=b)} = 1$ as should be expected

for $b = 2a \rightarrow P_{(-\infty, a=\frac{b}{2})} = \frac{a}{b} = \frac{1}{2}$

(e) $$\langle x \rangle = \int_{-\infty}^{\infty} x\,|\psi(x)|^2\,dx$$

$$= \int_{0}^{a} x\,\frac{A^2 x^2}{a^2}\,dx + \int_{a}^{b} x\,A^2\frac{(b-x)^2}{(b-a)^2}\,dx$$

$$= \frac{3}{b}\left[\frac{a^2}{4} + \left(-\frac{a^2}{4} + \frac{ab}{6} + \frac{b^2}{12}\right)\right]$$

$$= \frac{a}{2} + \frac{b}{4}$$

$\boxed{\text{Note}}$: $$\int_{x_0}^{x_1} (x-a)^n\,dx = \frac{(x-a)^{n+1}}{(n+1)}\Bigg|_{x_0}^{x_1}$$

the integral: $$\int_{x_0}^{x_1} x\,(b-x)^2\,dx = \int_{x_0}^{x_1} x\,(b^2 - 2bx + x^2)\,dx$$
$$= \left(\frac{b^2x^2}{2} - \frac{2bx^3}{3} + \frac{x^4}{4}\right)\Bigg|_{x_0}^{x_1}$$

or $\rightarrow$ $$\int_{x_0}^{x} (x-b+b)\,(b-x)^2\,dx$$

$$= -\int_{x_0}^{x_1} (b-x)^3\,dx + b\int_{x_0}^{x_1} (b-x)^2\,dx$$

$$= \left[+\frac{(x-b)^4}{4} + b\,\frac{(x-b)^3}{3}\right]\Bigg|_{x_0}^{x_1}$$

$$= \left[\frac{b^2x^2}{2} - \frac{2bx^3}{3} + \frac{x^4}{4}\right]\Bigg|_{x_0}^{x_1}$$

$$\psi(x,t) = A e^{-\lambda|x|} e^{-i\omega t}$$

P1.5

(a) $$1 = \int_{-\infty}^{\infty} dx\, |\psi(x,t)|^2 = A^2 \underbrace{|e^{-i\omega t}|^2}_{1} \int_{-\infty}^{\infty} e^{-2\lambda|x|} dx$$

$$= A^2 \cdot 2 \int_0^{\infty} dx\, e^{-2\lambda x} = \frac{2A^2}{2\lambda} = 1 \Rightarrow A = \sqrt{\lambda}$$

(b) $$\langle x \rangle = \lambda \int_{-\infty}^{\infty} x \left| e^{-\lambda|x|} e^{-i\omega t} \right|^2 dx =$$

$$= \lambda \int_{-\infty}^{\infty} dx\, x\, e^{-2\lambda|x|} = 0 \quad \text{by oddness.}$$

$$\langle x^2 \rangle = \lambda \int_{-\infty}^{\infty} dx\, x^2 e^{-2\lambda|x|} =$$

$$= 2\lambda \int_0^{\infty} dx\, x^2 e^{-2\lambda x}$$

To calculate $I_\lambda = \int_0^{\infty} dx\, x^2 e^{-2\lambda x}$ note that

$$I_\lambda = \frac{1}{4} \frac{d^2}{d\lambda^2} \int_0^{\infty} dx\, e^{-2\lambda x} = \frac{1}{4} \int_0^{\infty} dx\, (-2x)^2 e^{-2\lambda x}$$

Thus $$I_\lambda = \frac{1}{4} \frac{d^2}{d\lambda^2} \frac{1}{2\lambda} = \frac{1}{8} \frac{d}{d\lambda} \frac{-1}{\lambda^2} = \frac{1}{4} \frac{1}{\lambda^3}$$

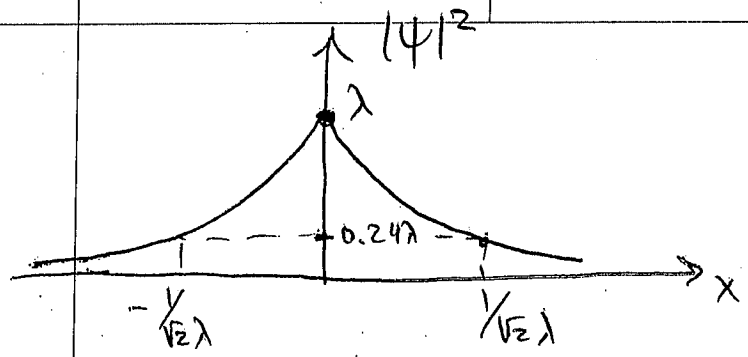
$$\Rightarrow \langle x^2 \rangle = 2\lambda \cdot \frac{1}{4\lambda^3} = \frac{1}{2\lambda^2}$$

(c) $$\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{1}{2\lambda^2} - 0 = \frac{1}{2\lambda^2}, \quad \sigma = \frac{1}{\sqrt{2}\lambda}$$

$$|\psi(x,t)|^2 = A^2 e^{-2\lambda|x|} = \lambda e^{-2\lambda|x|}$$

$$|\psi(x = \pm\sigma, t)|^2 = \lambda e^{-2\lambda/\sqrt{2}\lambda} = \lambda e^{-\sqrt{2}} \approx 0.24\lambda$$

$$|\psi(x=0,t)|^2 = \lambda$$



Probability to be outside $(-\sigma, +\sigma)$ interval is

$$P = \int_{-\infty}^{-\sigma} |\psi|^2 dx + \int_{\sigma}^{\infty} |\psi|^2 dx =$$

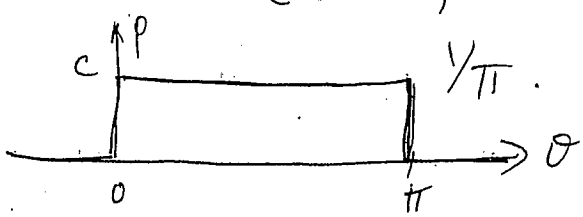
$$= 2\int_{\sigma}^{\infty} dx\ \lambda e^{-2\lambda|x|} = -e^{-2\lambda x}\Big|_{\sigma}^{\infty} = e^{-2\lambda\sigma} =$$

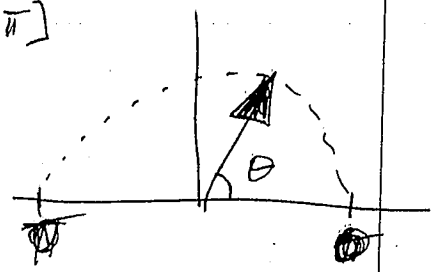
$$= e^{-\sqrt{2}} \approx 0.24$$

P1.11 Speedometer needle is free to swing and is equally likely to come to rest at any angle θ between 0 and π.

(a) Find $p(\theta)$.

$$p(\theta) = \begin{cases} \text{const} = C, & \text{for } \theta \in [0, \pi] \\ 0, & \text{otherwise} \end{cases}$$





We find C via: $\int_0^\pi p(\theta)\, d\theta = 1 \Rightarrow C \cdot \pi = 1$

$C = 1/\pi$

(b) $\langle \theta \rangle = \int_0^\pi \theta \frac{1}{\pi} d\theta = \frac{1}{\pi} \frac{\theta^2}{2}\Big|_0^\pi = \frac{\pi^2}{2\pi} = \frac{\pi}{2}$ (naturally)

$\langle \theta^2 \rangle = \frac{1}{\pi} \int \theta^2 d\theta = \frac{1}{\pi} \frac{\theta^3}{3}\Big|_0^\pi = \frac{\pi^2}{3}$

$\sigma^2 = \frac{\pi^2}{3} - \left(\frac{\pi}{2}\right)^2 = \pi^2 \left(\frac{1}{3} - \frac{1}{4}\right) = \pi^2 \frac{4-3}{12} = \frac{\pi^2}{12}$

$\Rightarrow \sigma = \pi/\sqrt{12} = \frac{\pi}{2\sqrt{3}}$.

(c) $\langle \sin\theta \rangle = \frac{1}{\pi} \int_0^\pi \sin\theta\, d\theta = \frac{1}{\pi} (-\cos\theta)\Big|_0^\pi = \frac{1}{\pi}(1-(-1)) = \frac{2}{\pi}$

$\langle \cos\theta \rangle = \frac{1}{\pi} \int \cos\theta\, d\theta = \frac{1}{\pi} \sin\theta\Big|_0^\pi = 0$

Note: $\langle \sin\theta \rangle = y/r$ $\quad \langle \cos\theta \rangle = x/r \Rightarrow \langle x \rangle = 0$ OK, $\langle y \rangle \neq 0$ OK

$\langle \cos^2\theta \rangle = \frac{1}{\pi} \int_0^\pi \cos^2\theta\, d\theta = \frac{1}{\pi} \frac{\pi}{2} = \frac{1}{2}$.

because $\cos^2\theta = \left(\frac{1}{2}(e^{i\theta} + e^{-i\theta})\right)^2 = \frac{1}{4}(e^{i2\theta} + e^{-i2\theta} + 2) = \frac{1}{2}(1 + \cos 2\theta)$